Calcurion Physics Worksheets

A-Level Physics

Topic: Thermal Physics

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. Define internal energy. **[2]**

2. State the first law of thermodynamics. **[2]**

3. What is meant by an ideal gas? **[2]**

4. State the ideal gas equation. **[1]**

5. Calculate the pressure of 0.20 mol of ideal gas at 300 K in a volume of 0.0050 m^3. ($R = 8.31$ J mol^{-1} K^{-1}) **[3]**

6. State Boyle's law. **[1]**

7. A gas at 100 kPa and 0.2 m^3 is compressed to 0.05 m^3 at constant temperature. Calculate the new pressure. **[2]**

8. Define specific heat capacity. **[1]**

9. 0.50 kg of water is heated from 20°C to 80°C. $c = 4200$ J kg^{-1} K^{-1}. Calculate energy required. **[3]**

10. Define specific latent heat of fusion. **[1]**

11. Calculate energy to melt 0.20 kg of ice at 0°C. $L_f = 3.34 \times 10^5$ J kg^{-1}. **[2]**

12. State the relationship between absolute temperature and average kinetic energy of gas molecules. **[2]**

13. Calculate the r.m.s. speed of oxygen molecules at 300 K. ($M = 0.032$ kg mol^{-1}, $R = 8.31$) **[3]**

14. What is Brownian motion and what does it provide evidence for? **[2]**

15. Sketch a p–V graph for an isothermal expansion of an ideal gas. **[2]**

Section B – Hard (Q16–25)

16. Derive $pV = \frac{1}{3}Nm\langle c^2\rangle$ from kinetic theory assumptions (outline). **[4]**

17. Show that $\frac{1}{2}m\langle c^2\rangle = \frac{3}{2}kT$. **[3]**

18. 2.0 mol of ideal gas expands isothermally from 0.01 m^3 to 0.04 m^3 at 300 K. Calculate the work done by the gas. **[4]**

19. In an adiabatic process, what happens to the temperature of an expanding gas and why? **[3]**

20. A gas is heated at constant volume. Explain the energy transfers using the first law. **[3]**

21. Calculate the change in internal energy when 500 J of heat is added and 200 J of work is done by the system. **[2]**

22. Explain why the molar heat capacity at constant pressure is greater than at constant volume. **[3]**

23. A metal block of mass 0.40 kg and $c = 400$ J kg^{-1} K^{-1} cools from 80°C to 20°C. Calculate energy released. **[2]**

24. Describe an experiment to measure the specific heat capacity of a liquid using the electrical method. **[4]**

25. Explain the concept of absolute zero in terms of molecular kinetic energy. **[2]**

End of Worksheet • *Show all working clearly.*